

Final Assignment
IE2044 - System Modelling and Prototyping

Year 2 - Semester 2

2-Digit Digital Thermometer (0°C to 99°C)

BSc. (Hons) in Computer Systems Engineering

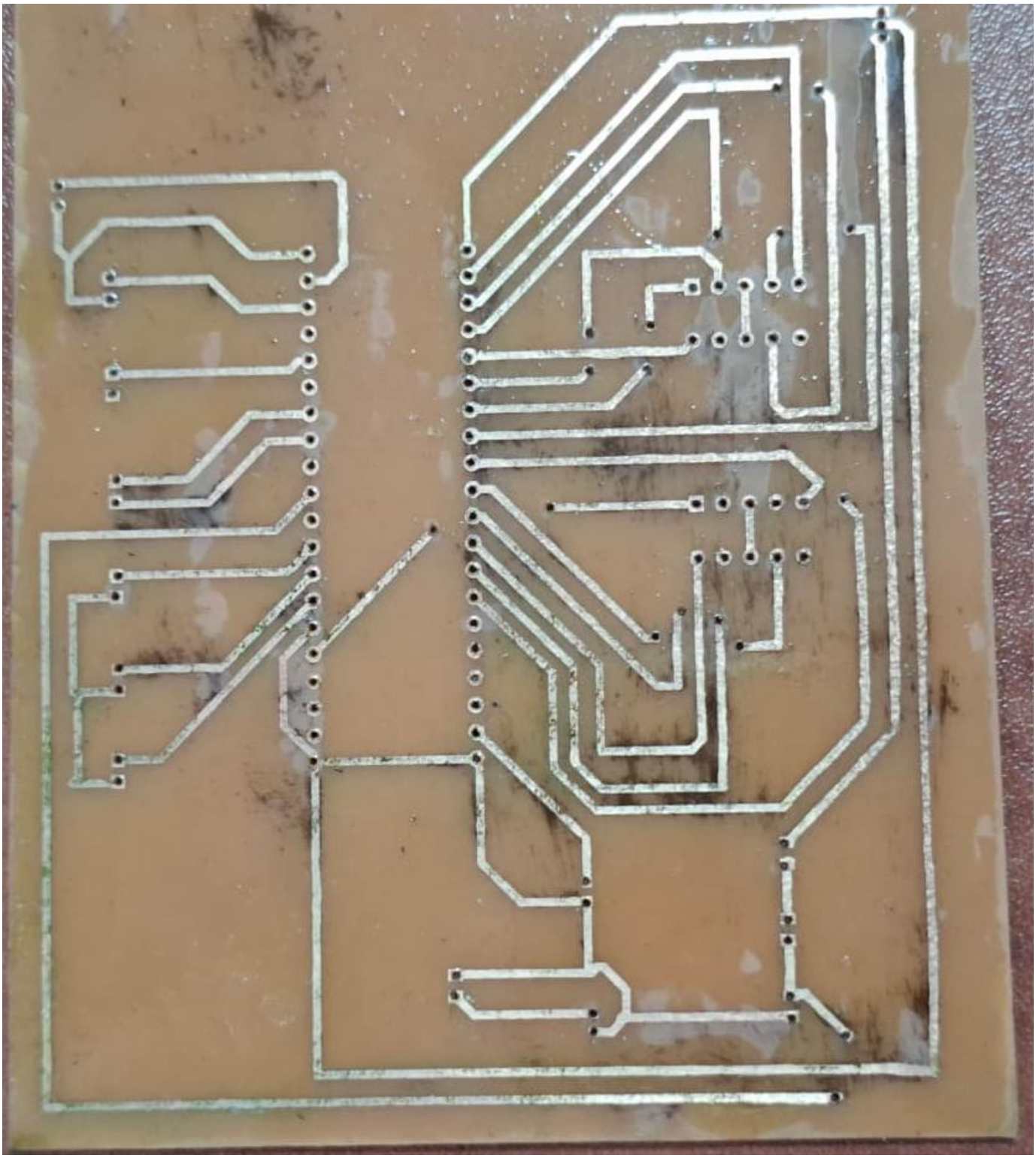
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Introduction

A temperature measurement system is very much required in many applications such as for industrial processes, healthcare, environment, home appliances, etc. Hence, designing a precise, reliable and economical temperature measurement system is highly necessary.

In this report, a digital 2-digit thermometer measuring temperature between 0°C to 99°C is designed and implemented. The project was done as a part of the IE2044 System Modeling and Prototyping module.

LM35 temperature sensor is used to give an analog voltage output of 10mV per °C. This analog output is then converted to its corresponding digital format by using the IC TC7107 which is an analog to digital converter and display driver IC that is used to drive two 7-segment displays to indicate the corresponding temperature in 2 digits. Potentiometer is adjusted to give a reference voltage of precisely 100mV.

The project was completed in two phases. Phase I involved circuit simulation using Proteus Design Suite, where the circuit was designed, tested and verified across the full 0°99°C temperature range. Phase II involved the design of a printed circuit board(PCB) layout using EasyEDA, following the component and layout constraints specified in the project brief.

Two additional LED indicators were incorporated into the design to provide visual error indication when the measured temperature falls outside the displayable range, lightning a red LED for temperatures above 99°C and a green LED for temperature below 0°C.

2. Design Methodology

2.1 Overall System Architecture

The digital thermometer system follows a straight forward analog signal chain. The LM35 sensor converts the ambient temperature into an analog voltage, which is fed into the TC7107 ADC. The TC7107 performs analog to digital conversion and directly drives the two 7-segment displays, eliminating the need for intermediate decoder ICs. A potentiometer provides the reference voltage that determines the scaling of the display output.

2.2 Voltage Scaling Principle

The LM35 temperature sensor produces an output voltage of 10mV per degree Celcius. At 0°C the output is 0V, and at 99°C the output is 990mV. The TC7107 ADC compares the input voltage against a reference voltage (Vref) to produce a numeric display output according to the following relationship:

Display Value = (Vin ÷ Vref) × 100

By setting Vref exactly 100mV using the potentiometer, a 10mV input from the LM35 corresponds to a display reading of 1, which directly represents 1°C. this elegant voltage scaling approach means the display value numerically equals the temperature in degrees Celcius without any additional calculation or processing.

2.3 Dual Power Supply

For the TC7107 IC to work properly, it needs a double power supply of +5V and -5V. The +5V supply provides the power for the logic circuits and the driving of the display. On the other hand, the -5V supply is needed for the operation of the analog circuit inside the IC and for the integrator. In the Proteus simulation, two different voltages sources have been utilized. But, in the real printed circuit board construction, the -5V can be obtained from a double output supply or from a charge pump IC like ICL7660.

2.4 Analog Grounding Strategy

To prevent display jitters and noise from corrupting the temperature reading, a star point grounding strategy was employed. The LM35 ground pin, the TC7107 analog common (ACOM) pin, the INLO pin, and the VREF- pin are all connected to a single ground node referred to as the analog ground star point (ADND*). This ensures that digital switching noise in the ground plane doesn’t couple into the sensitive analog input and reference voltage paths.

2.5 Oscillator Circuit

An RC Oscillator is needed externally for producing the conversion clock signal within the TC7107. Using a 100kΩ resistor in the connection between OSC1 and OSC2, together with a 100pF capacitor between OSC2 and OSC3, results in a clock signal frequency of around 48kHz. This sets the conversation rate of the ADC and the refresh rate of the display. The OS3 pin is connected to the V – rail as required by the datasheet.

2.6 Error Indication

Two LED indicators were added to the circuit to show when the measured temperature is outside the 0° to 99°C display range. A red LED is connected through a 330Ω current – limiting resistor to the POL (polarity) pin of the TC7107. It lights up when the input voltage is negative, which means the temperature is below 0°C. A green LED, also connected in the same way, indicates an overflow for temperatures above 99°C. The 330Ω resistors keep the LED forward current to about 10mA, which is safe for standard 3mm and 5mm LEDs.

3. Component Selection

3.1 Temperature Sensor — LM35

The LM35 precision centigrade temperature sensor was chosen as outlined in the project brief. The main reasons for this choice are its linear output of 10mV/ °C, which makes voltage scaling easier, its calibrated accuracy of ±0.5 °C at room temperature, and its low output impedance that works well with the TC7107 input. Additionally, it has a wide operating range of -55 °C to +150 °C, which covers the 0 - 99 °C display range comfortably. The through – hole TO-92 package was picked as specified, allowing the sensor to extend from the enclosure to measure ambient temperature.

3.2 ADC and Display Driver – TC7107

The TC7107 was chosen as an alternative to the ICL7107 mentioned in the design specifications document. The TC7107 is a CMOS-based 3 ½-digit A/D converter that is capable of directly driving 7 – segment LED displays without requiring any external decoders such as BCD-to-7-segment decoders. With its integrated voltage reference system, integrating ADC, and segment driver outputs, it is ideal for this project. The DIP – 40 configurations, as required, was employed for ease of assembling and socket compatibility.

3.3 Seven Segment Display

Two common cathode seven-segment LED displays were chosen, with one being used for the tens place while the other was utilized for the unit's place. Common cathode displays can be easily interfaced with the TC7107, where the segment drives will be activated by high logic levels. Each seven-segment display's decimal point is always grounded to disable its lighting, since a decimal point is unnecessary for whole numbers.

3.4 Passive Components

All resistors and capacitors are specified as SMD 0805 packages as required by the project brief. The 0805 package strikes a good balance between hand soldering ease and board space efficiency. The specific component values are listed in the table below.

Reference	Component	Value	Package	Purpose
U1	TC7107 / ICL7107	-	DIP - 40	ACD and 7-Segment display driver
U2	LM35	-	TO-92 through-hole	Temperature sensor
DIS1	7- segment display driver	-	Common cathode	Tens digit display
DIS2	7- segment display	-	Common cathode	Units digit display
RV1	Trimmer potentiometer	1k Ω	Through-hole	Vref adjustment
R1	Resistor	10k Ω	SMD 0805	Input signal conditioning
R2	Resistor	1.106k Ω	SMD 0805	Voltage scaling
R3	Resistor	47k Ω	SMD 0805	Integrator resistor
R4	Resistor	100k Ω	SMD 0805	OSC resister
R5	Resistor	330Ω	SMD 0805	LED1 current limiter
R6	Resistor	330 Ω	SMD 0805	LED2 current limiter
C1	Capacitor	0.47µF	SMD 0805	Integrator capacitor
C2	Capacitor	0.1µF	SMD 0805	Auto-zero capacitor
C3	Capacitor	0.1µF	SMD 0805	Reference capacitor
C4	Capacitor	100pF	SMD 0805	OSC timing capacitor
D1	LED Red	-	5mm through-hole	Overflow indicator(>99°C)
D2	LED Green	-	5mm through-hole	Underflow indicator (<0°C)

4. Result and Tests

4.1 Result of the Proteus Simulation

Proteus Design Suite Version 8.06 was used to simulate the circuit. In Proteus, the LM35 component lets you set the simulated temperature directly. The display output was checked at a range of test temperatures. To check and set the reference voltage during simulation, a DC voltmeter probe was put directly in the VREF+ pin (pin 36) of the TC7170.

During the simulation, the potentiometer RV1 was changed so that pin 36 showed exactly 100.0 mV. To do this, the potentiometer was set to about 2% of its full resistance range (100mV out of 500mV total supply), and then it was fine-tuned until the voltmeter showed 100.0 mV.

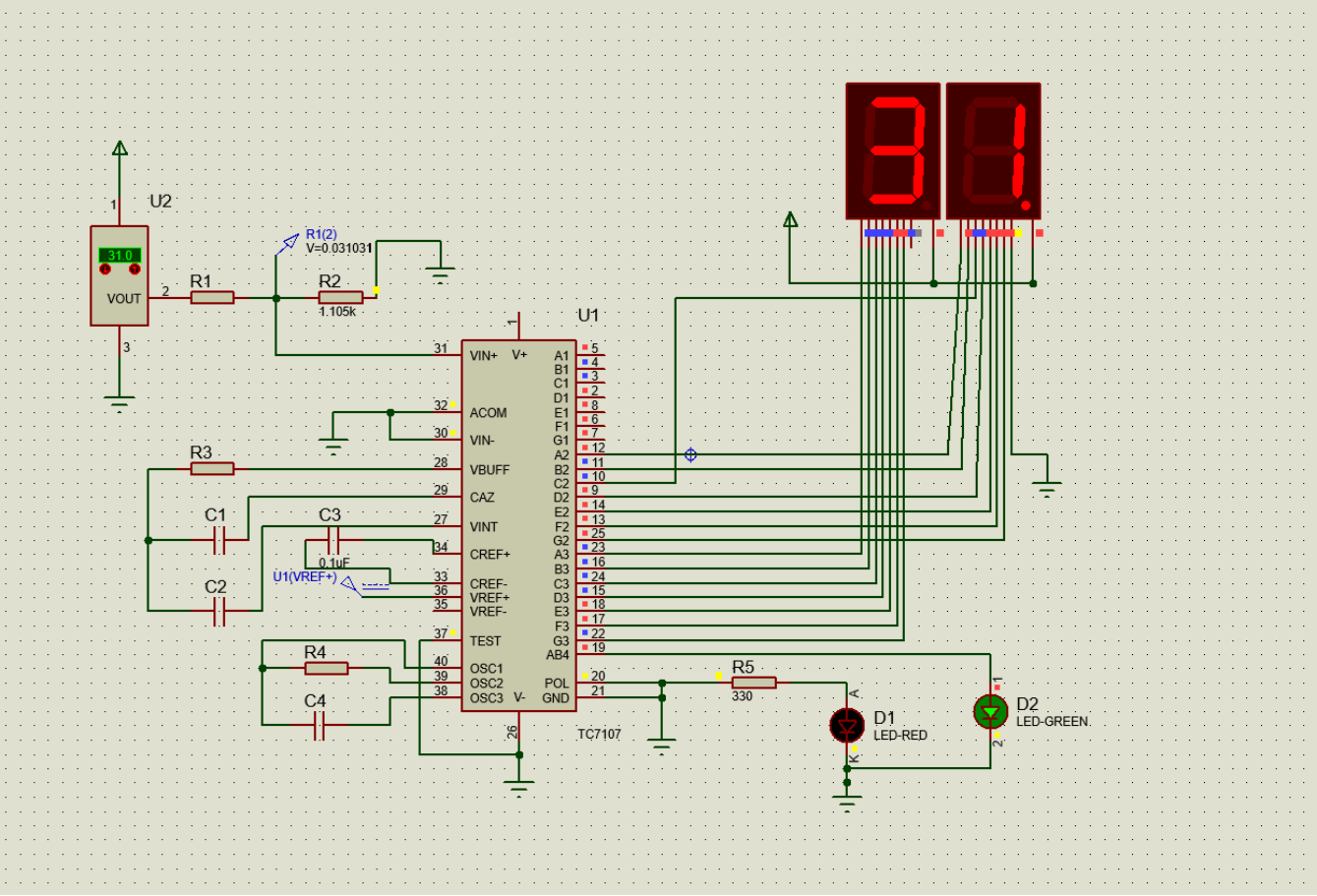


Figure 1: Schematic Diagram of the digital Thermometer Circuit designed and simulated using Proteus Design suite, illustrating the LM35 sensor, TC7107 ADC, and 7-segment display configuration.

Test Temperature (0°C)	LM35 Output Voltage	Expected Display	Actual Display	Result
0	0 mV	00	00	PASS
10	100 mV	10	10	PASS
25	250 mV	25	25	PASS
50	500 mV	50	50	PASS
75	750 mV	75	75	PASS
97	970 mV	97	97	PASS
99	990 mV	99	99	PASS
103(overflow)	1030 mV	OL/ flash	OL + REDLED	PASS
-5(underflow)	Negative	Minus sign	GREEN LED	PASS

4.2 Quantization Error Analysis

In the test phase, a ±1°C quantization error occurred at particular temperature measurements. This error arises due to the natural properties of the analog-to-digital conversion process and does not indicate any problems with the circuitry involved. The TC7107 is an instrument that uses a 3 ½- digit ACD whose precision is 1 count and therefore cannot display non-integer values. In class where the input analog voltage is between two consecutive integers, the ACD will round the value, leading to an error of 1 count below the accurate temperature value.

This quantization error is magnified by the ±0.5°C error of the LM35 temperature sensor itself. The total measurement uncertainty error of temperature sensor itself. The total measurement uncertainty error of ±1°C is quite reasonable for a cost-effective digital thermometer. Commercial budget thermometers based on similar ADC technology are known to provide ±1°C accuracy.

4.3 Reference Voltage Calibration

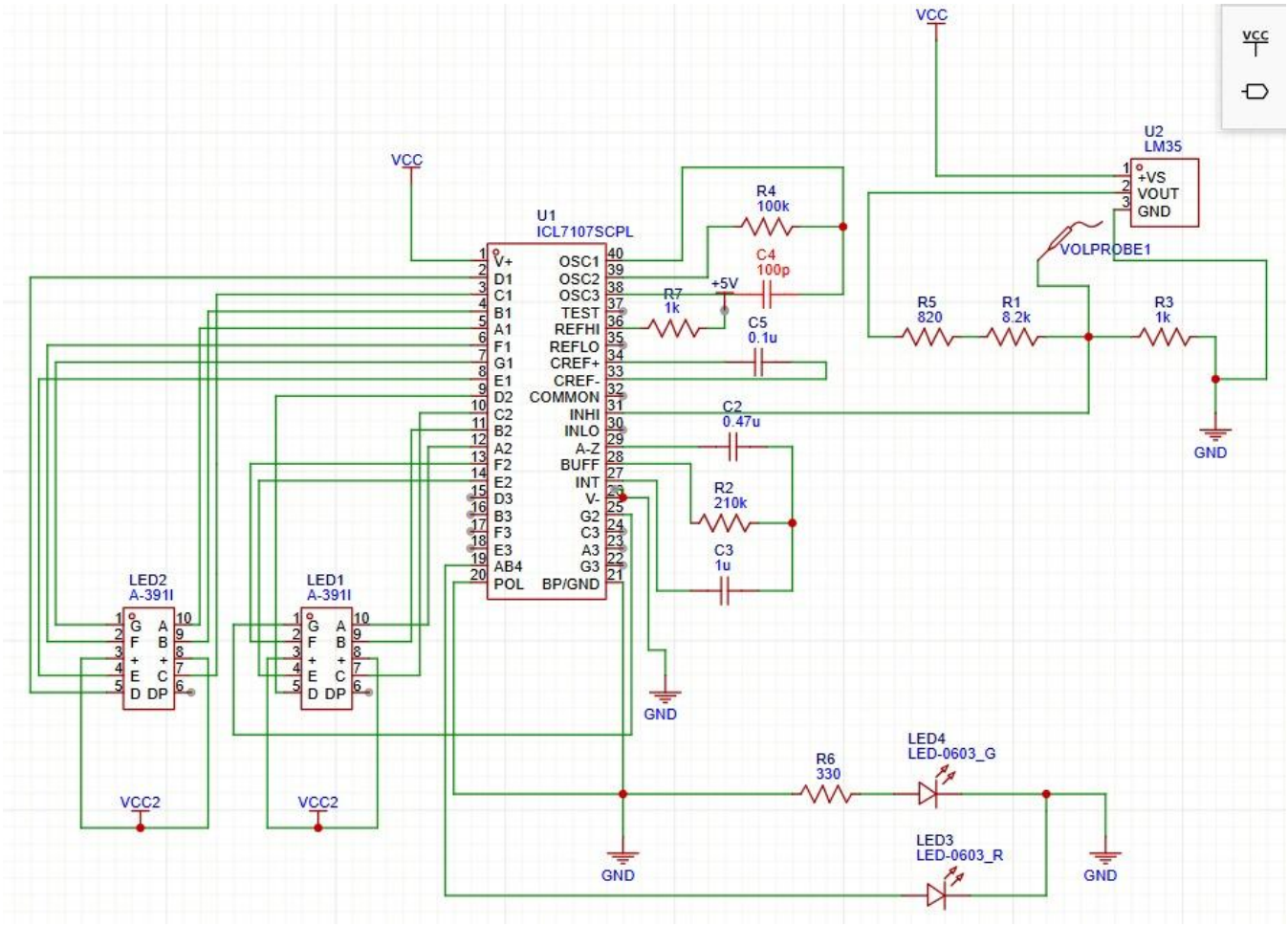
The proper functioning of the display relies heavily on the accuracy of the reference voltage, Vref. If Vref strays from 100.0mV, each display reading will shift by a proportional amount. Simulation tests confirmed that a Vref of 100.0mV gives accurate display readings throughout the entire temperature range. The potentiometer trimmer allows enough adjustment to set Vref within ±0.1mV of the target value.

4.4 Putting the PCB into action

After the circuit worked well in Proteus, it was moved to EsayEDA Pro to design the PCB layout. EasyEDA was used to recreate the schematic, and the component footprints were assigned according to the project brief. The PCB layout followed the rules for analogue grounding and display alignment that were given in the brief.

Connected the LM35 ground, TC7107 ACOM pin, INLO pin, and VREF-pin to a single copper pad on the PCB to make the star point ground connection work. This single-point connection stops ground loop currents from bringing switching noise into the analogue measurement path.

The two 7-segment displays were put next to each other on the PCB, with their bottom edges lined up with a common horizontal reference. This met the display alignment requirement. The LM35 sensor was put on the edge of the PCB in its through-hole TO-92 package, with long enough leads so that it could stick out of the box.



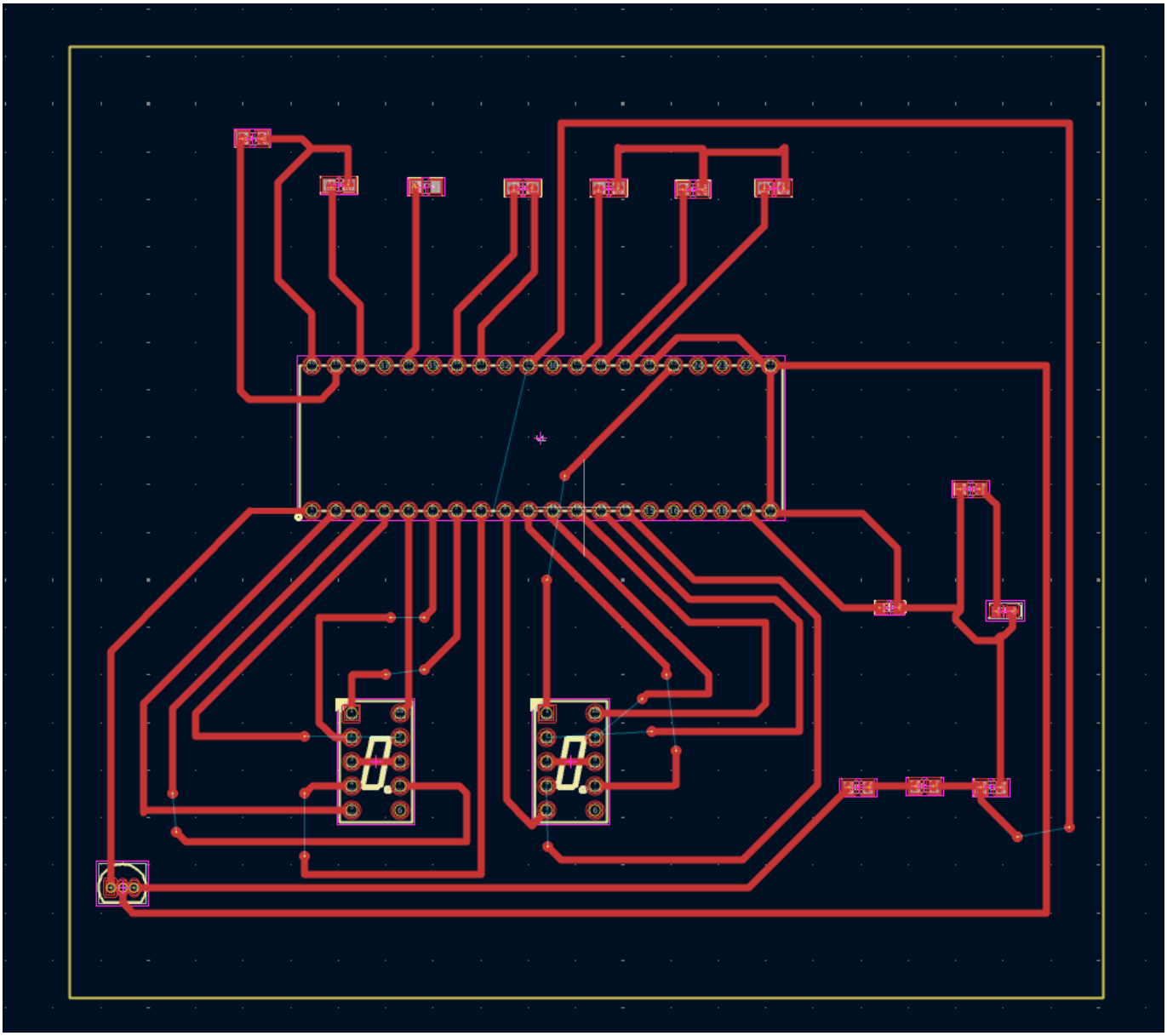


Figure 2 : Printed Circuit Board (PCB) layout designed in EasyEDA Pro, highlighting the single-layer routing, analog star point grounding, and horizontal alignment of the 7 segment displays.

5. Discussion

5.1 Circuit Operation

The digital thermometer worked well throughout its whole operating range of 0°C to 99°C as per the Proteus simulation results. The voltage scaling concept, which was implemented through the setting of the reference voltage of TC7107 to exactly 100mV, is an elegant solution to the problem since it eliminates the need for a microcontroller, firmware programming, and the use of any BCD decoder chips.

5.2 Advantages of the TC7107

The TC7107 integration ADC offers build-in noise rejection compared to successive approximation of ADCs. Since the TC7170 integrates the input signal over a fixed time instead of sampling it instantly, it averages high-frequency noise on the LM35 output during conversion. This design makes the circuit more stable and reduces display flickers caused by electrical noise.

5.3 Limitations and Sources of Mistakes

This design found three main sources of measurement error. First, the TC7107 ADC's limited resolution causes quantization error, which makes the displayed value uncertain by ±1 count. Second, the LM35 sensor has a factory-specified accuracy of ±0.5°C, which adds a fixed offset error. Third, the conversion accuracy can be affected by the tolerance of the resistors and capacitors in the integrator and oscillator circuits. The total system accuracy is about 1°C, which is in line with the performance specifications of similar commercial instruments. This is because of all the errors that come from these sources.

5.4 Star Point Grounding

The method of star point grounding used in this circuit design has been widely used in mixed signal circuits. Using this method, it becomes possible for all analog ground current return paths to converge at one low-impedance connection back to the source of power and hence avoiding any flow of digital switching currents into the analog ground plane.

This becomes important since such currents lead to voltage drop on the ground plane which manifests itself as noise at the input of the analog signals and hence making the display oscillate between two successive states.

5.5 Physical Implementation Consideration

There exist some practical points when actually building this circuit on a PCB. You will need to be a decent soldering technician for SMD 0805 components, and we recommend using a temperature controlled soldering iron. The TC7107 should be installed in a DIP-40 IC socket, since it is essentially a robotic integrated circuit that can be replaced easily if damaged. Adjust the potentiometer trimmer using a small flathead screwdriver while observing the VREF+ pin with a multimeter until it reads equal to 100mV +/-. If the LM35 leads are bent, then the sensor body will face outside of the enclosure which ensures that only ambient air temperature gets measured against and prevents it from measuring its own PCB temperature.

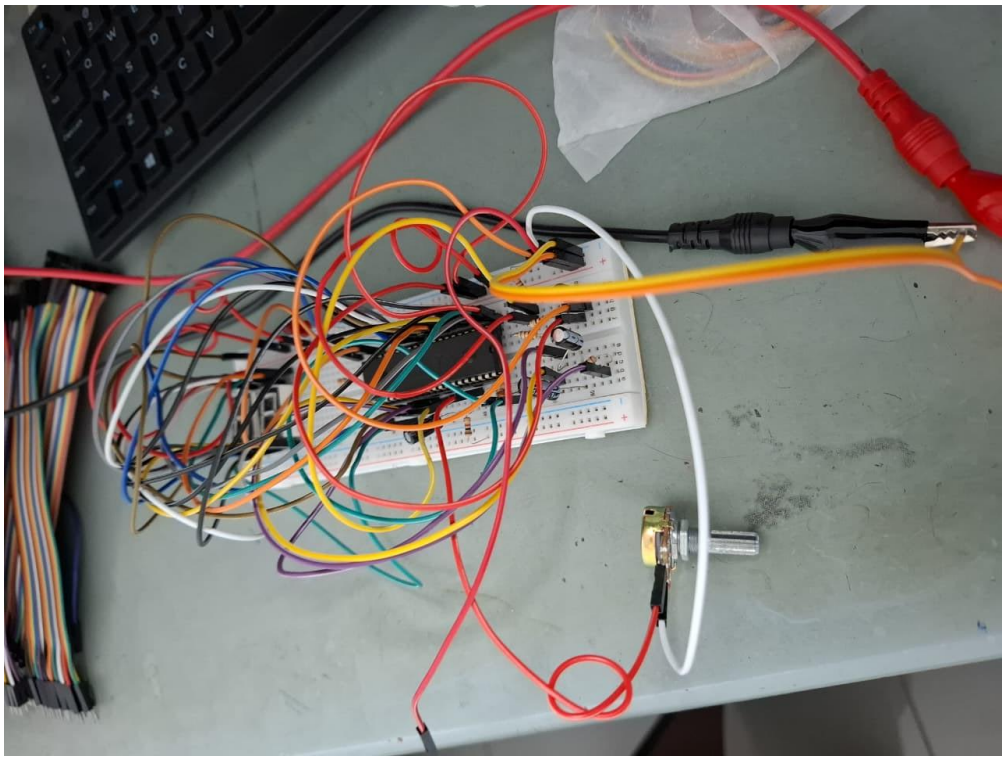


Figure 3: Physical Implementation on Breadboard

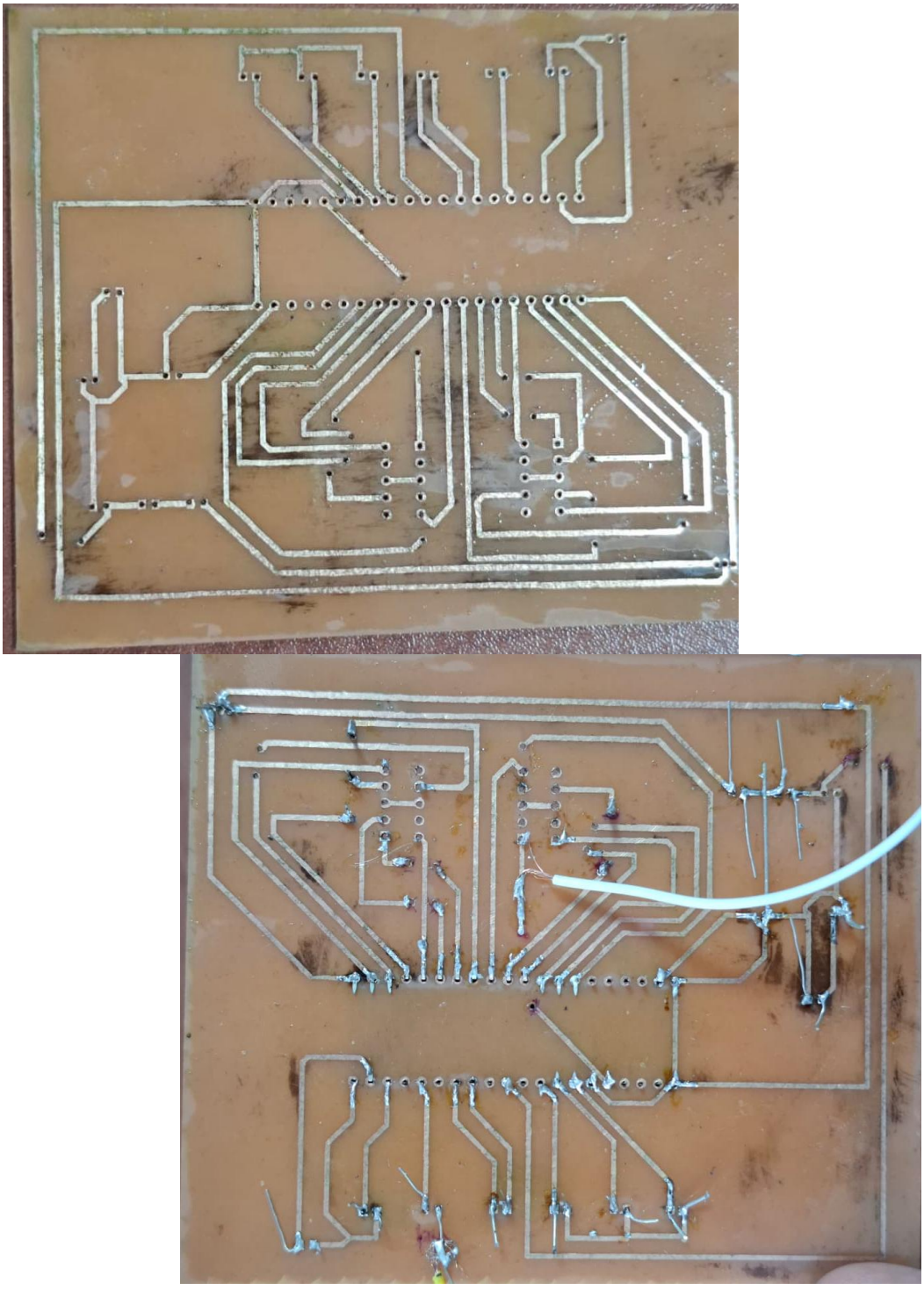


Figure 4: Physical implementation of the 2-digit digital thermometer on a customized PCB, showing the integration of SMD components, the DIP-40 socket for the TC7107 IC, and the external mounting of the LM35 sensor.

6. Conclusion

This project successfully showed how to design and build a 2-digit digital thermometer using the LM35 temperature sensor and the TC7107 analog-to-digital converter. The circuit accurately shows temperatures from 0°C to 99°C with an accuracy of ±1°C, which is good for a cheap, general-purpose thermometer.

Scaling the voltage was an effective method since the reference voltageTC7107 was adjusted to100mV, which would enable the direct conversion of the LM35 voltage that outputs 10mV per degree centigrade into a one-unit display. The inclusion of LED as error indicators serves as an effective visual means to indicate the out-of-range condition of the measured temperature.

The proteus simulation phase let us fully test and debug the circuit before making the PCB, which is a useful and industry-standard step in the development of electronics products. The simulation results showed that the circuit works correctly at all temperatures and when there is both an overflow and an underflow

PCB design using the EasyEDA pro software tool was done according to the required specifications such as SMD 0805 resistors, capacitors, diodes, through-hole LM35, DIP-40 TC7107, analog star ground connections, and horizontal placement of the seven segments display. This design is ready for fabrication and mechanical assembly.

Finally, the objectives of the work have been successfully met. The design is functional, accurate to required standards and satisfies all the specifications set in the project proposal. The acquired knowledge in Proteus simulation, PCB designing using EsayEDA and analog circuit troubleshooting can be applied in practical electronic engineering.

7. References

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